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Speed Up Similarity Search of Time Series Under Dynamic Time Warping

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ABSTRACT Similarity search is a foundational task in time series data mining. Although there are many ways to measure the similarity of time series, a lot of evidence indicates that dynamic time warping (DTW) has the best robustness in many applications. Unfortunately, the expensive computational cost limits its application in large-scale databases. To speed up similarity search under DTW, we design a framework of two-stage similarity search for time series. In the first stage, we propose an improved lower bounding distance, which can be used to discard plenty of dissimilar series to get a set of candidate sequences. In the second stage, to efficiently get the retrieval result from the set of candidate sequences, we explore early abandoning strategy to avoid the full calculation of DTW. Extensive experiments are conducted on real-world data sets. The experimental results indicate that the proposed method can improve the retrieval efficiency of similarity search under DTW and guarantee no false dismissals.

INDEX TERMS Time series, similarity search, dynamic time warping, lower bounding distance, early abandoning strategy.

I. INTRODUCTION

Time series exist in almost every field of human activities, such as economy [1], politics [2], medicine [3], traffic [4], and multimedia [5]. With the explosive increase of data storage, data mining tasks for time series, such as classification [6], clustering [7], outlier detection [8], and similarity search [9], have been attracting great attention.

Similarity search is to find all the series that are similar to the query sequence. Similarity measure is used to judge whether two time series are similar or not. The popular similarity measures of time series are Euclidean distance and dynamic time warping (DTW). Euclidean distance is parameter-free and has linear computational complexity. But it is sensitive to various deformation of time series [10]. DTW searches an optimal match between two sequences, making it applicable to accommodate the deformation of time series. Thus, it can obtain high matching accuracy [11].

Although there are many ways to measure the similarity of time series, there is growing evidence that DTW is the best

similarity measure in most applications [12]. Unfortunately, the computational cost of DTW is expensive, which restricts its applications in similarity search on large-scale data sets.

In this paper, we propose a two-stage similarity search to alleviate the contradiction between computational cost and retrieval efficiency. The main contributions of this paper are summarized as follows.

- A simple and effective framework of similarity search for time series is proposed, which combines an improved lower bounding distance and early abandoning strategy together.
- An improved lower bounding distance is proposed. The relevant proof is completed to show it can guarantee no false dismissals.
- Extensive experiments are conducted on real-world data sets to verify the effectiveness of our proposed method.

The rest of the paper is organized as follows. In Section II, we formulate the problem of similarity search for time series and give a brief review of the related work. In Section III, we propose a lower bounding distance of DTW and design a framework of two-stage similarity search. In Section IV, we conduct extensive experiments on real-world data sets,

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Algorithm 1 Sequential Scan**Require:** Q, S, ϵ ;**Ensure:** R ;

- 1: forecast sequence S_i in a time series data set S
- 2: if $D_{dtw}(Q, S_i) \leq \epsilon$
- 3: add S_i to R ;
- 4: end
- 5: return R

and evaluate the effectiveness of the proposed method. Finally, we summarize the conclusion in Section V and present future directions.

II. BACKGROUND**A. PROBLEM FORMULATION**

Time series is a sequence of observations $x(t)$ ($t = 1, 2, \dots, n$), where n is the length of the series [13]. Since time series from a data set usually have the same interval, they can be denoted as $X = x_1, x_2, \dots, x_n$.

Definition 1: DTW distance between two time series $X = x_1, x_2, \dots, x_n$ and $Y = y_1, y_2, \dots, y_m$ is defined as [14]:

$$D_{dtw}(X, Y) = D_{base}(x_1, y_1) + \min \begin{cases} D_{dtw}(X, Y[2:-]) \\ D_{dtw}(X[2:-], Y) \\ D_{dtw}(X[2:-], Y[2:-]) \end{cases} \quad (1)$$

where $D_{base}(x_i, y_j)$ is the base distance:

$$D_{base}(x_i, y_j) = |x_i - y_j| \quad (2)$$

Definition 2: The similarity search in this paper refers to the whole sequence range search, which can be described as follows. Given a time series data set $S = S_1, \dots, S_k$, a query sequence Q and a distance threshold ϵ , we want to find all the time series S_i from S , such that $D_{dtw}(Q, S_i) \leq \epsilon$, where $1 \leq i \leq k$.

B. RELATED WORK

The basic approach of similarity search is sequential scan, as shown in Algorithm 1. It calculates the DTW distance between the query sequence Q and every time series S_i in the data set S .

Due to the expensive computational cost of DTW, sequential scan cannot scale up to large data sets. To improve retrieval efficiency, some hardware-based speedup techniques were proposed [15]. They utilized GPUs and FPGAs to accelerate similarity search under DTW. But most methods focused on software-based solutions. These methods can be divided into two categories: similarity search with indexes and without indexes.

1) SIMILARITY SEARCH WITH INDEXES

In order to apply DTW in large-scale data sets, various indexing methods were proposed. And they usually use the two-stage search strategy, as illustrated in Fig. 1.

In the first stage (index query stage), time series in the data set are mapped into geometric objects in the low dimensional space. Indexes for these low dimensional geometric objects are established. Since DTW dissatisfies triangle inequality, it cannot be directly used to retrieve similar sequences on the index. So the lower bounding distance of DTW is used to approximately search similar series on the index. Then we can get a candidate set, containing not only time series meeting the condition of similarity search, but also false alarms whose DTW distances with the query sequence are larger than ϵ . Therefore, this stage is called rough search.

In the second stage (post-processing stage), sequential scan is adopted to remove false alarms from the candidate set. Since plenty of dissimilar series are efficiently excluded in the index query stage, the number of the series in the candidate set is much smaller than that of the series in the original data set. Therefore, compared with sequential scan on the original data set, the two-stage similarity search is able to improve the retrieval efficiency.

The key points of these methods are their lower bounding distances of DTW. Yi *et al.* [16] and Keogh and Ratanamahatana [17] respectively proposed classic lower bounding distances, labelled as LB_Yi and LB_Keogh.

Given two time series, LB_Yi took one of the series as the reference series, and extracted some points of the other series to construct the lower bounding distance. The values of these points should be larger than the maximum value of the reference series, or smaller than the minimum value of the reference series.

In order to improve the tightness of lower bounding distances, Keogh proposed his lower bounding distance. LB_Keogh extracts the upper and lower boundary series of the query sequence, according to the global or local constraints of the bending path. LB_Keogh calculates Euclidean distance between the sequence in a data set and the region surrounded by the upper and lower boundary series.

Both LB_Yi and LB_Keogh can guarantee no false dismissals. The tightness of LB_Keogh is better than that of LB_Yi. So LB_Keogh can exclude more dissimilar series in the rough search stage. Furthermore, Zhu and Shasha [18], Zhou and Wong [19], etc. proposed their improvements on LB_Keogh respectively.

2) SIMILARITY SEARCH WITHOUT INDEXES

Since DTW dissatisfies triangle inequality, it is not easy to design indexes of time series for similarity search. Multidimensional index structures begin to degrade rapidly, somewhere above 16 dimensions [20], [21]. Besides, maintaining indexes will cost extra resources. Therefore, indexes are abandoned for non-large-scale data sets.

Rakthanmanon *et al.* [22] proposed the UCR Suite by integrating many speedup methods, which extends similarity search to an order of trillions of observations. Besides, Silva *et al.* [23], [24] proposed DTW with pruned warping paths (PrunedDTW) to improve the internal efficiency of the DTW calculation. It can be utilized to remove redundant

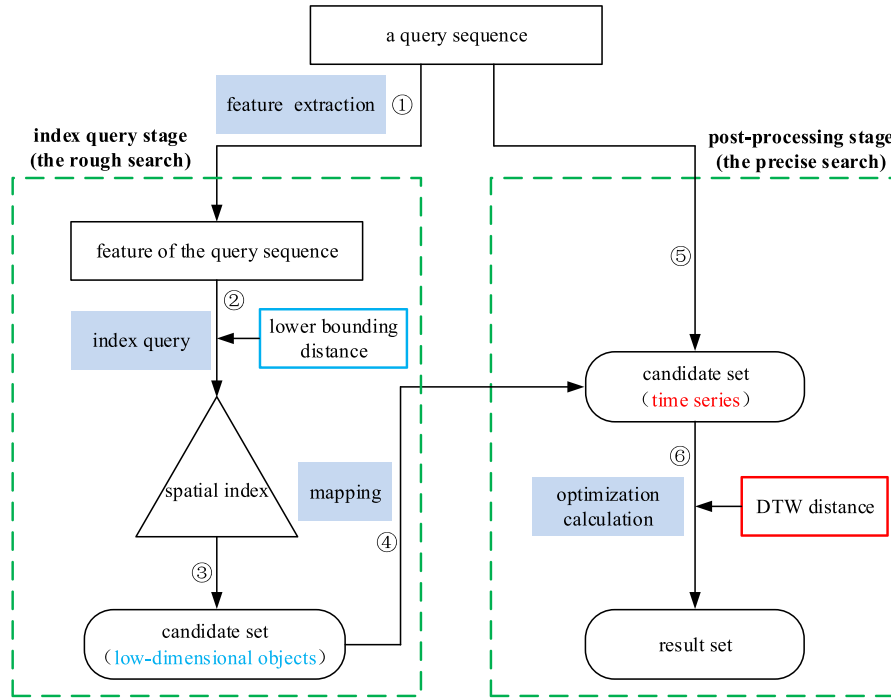


FIGURE 1. Procedures of the two-stage similarity search under DTW.

calculations of DTW to further speed up the UCR suite. However, the UCR Suite and PrunedDTW focus on the nearest neighbor subsequence search. They are not directly applicable to the whole sequence range search.

From this perspective, we try to propose a simple method to speed up the whole sequence range search. The proposed method can be applied to large-scale data sets without indexes, and the candidate sets in the post-processing stage of similarity search with indexes, as illustrated in Fig. 1.

III. PROPOSED METHOD

In this section, we design a framework of two-stage similarity search for time series. In the first stage, a lower bounding distance of DTW is proposed, which is used to discard plenty of dissimilar series efficiently. In the second stage, early abandoning strategy is utilized to avoid the full calculation of DTW.

We first introduce some basic notations that will be used throughout the paper, as shown in Table 1.

A. LOWER BOUNDING DISTANCE

1) THE WARPING PATH OF DYNAMIC TIME WARP

DTW is to achieve the alignment between $Q = q_1, q_2, \dots, q_n$ and $S_i = s_1, s_2, \dots, s_m$, as shown in Fig. 2(b). Each alignment corresponds to a warping path W , as shown in Fig. 2(c).

A warping path W consists of a set of matrix elements, defining the mapping between Q and S_i :

$$W = w_1, w_2, \dots, w_K$$

$$\max(n, m) \leq K < n + m - 1 \quad (3)$$

TABLE 1. The basic notations in this work.

S	Time series data set
C	Time series candidate set
R	Result set of similarity search
S_i	The i th sequence in the time series data set S
Q	A query sequence of length n , $Q = q_1, q_2, \dots, q_n$
$D_{dtw}(Q, S_i)$	The DTW distance between Q and S_i
DTW	The dynamic time warping distance
EA_DTW	Early abandoning strategy for the calculation of DTW
LB_Li	The lower bounding distance proposed in this work

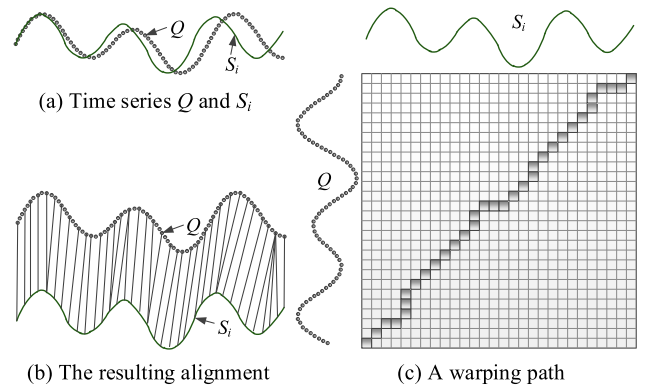


FIGURE 2. Warping path and resulting alignment.

The k -th element of W is denoted as $w_k = (i, j)_k$, which indicates the alignment between q_i and s_j . The warping path should satisfy the following constraints [17]:

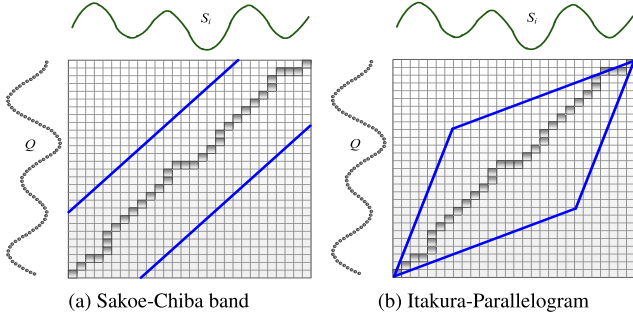


FIGURE 3. The region of the warping path.

Boundary conditions: $w_1 = (1, 1)$, $w_K = (n, m)$. The warping path starts from (q_1, s_1) and ends with (q_n, s_m) .

Continuity: Given $w_k = (a, b)$, $w_{k-1} = (a', b')$, then $a - a' \leq 1$ and $b - b' \leq 1$. The allowable steps in the warping path are restricted to adjacent cells.

Monotonicity: Given $w_k = (a, b)$, $w_{k-1} = (a', b')$, then $a - a' \geq 0$ and $b - b' \geq 0$. The elements of W increase monotonously in the time dimension.

There are several warping paths satisfying these conditions above. DTW corresponds to the path that minimizes the warping cost:

$$D_{dtw}(Q, S_i) = \min\{\sum_{k=1}^K d(w_k)\}. \quad (4)$$

where $d(w_k) = |q_i - s_j|$.

2) GLOBAL CONSTRAINTS ON TIME WARPING

In addition to the three constraints above, global constraints are usually introduced in many applications, which restricts warping paths to certain region called the warping window. Sakoe-Chiba band and Itakura-parallelogram [25] are two common global constraints, as illustrated in Fig. 3. Their warping windows are a band and a parallelogram in the diagonal direction.

Global constraints can prevent pathological warpings, where a small section of a sequence maps onto a large section of another sequence [17]. Without loss of generality, we mainly involve Sakoe-Chiba band in the paper.

3) PROPOSED LOWER BOUNDING DISTANCE

Under global constraints, the warping path $w_k = (i, j)_k$ is subject to $j - r \leq i \leq j + r$, where r is a term allowed range of warping. For a point in a sequence, only part of points of another sequence can be matched with it. Then, two sequences $U = u_1, u_2, \dots, u_n$ and $L = l_1, l_2, \dots, l_n$ are defined as:

$$u_i = \max\{q_{i-r} : q_{i+r}\} \quad (5)$$

$$l_i = \min\{q_{i-r} : q_{i+r}\} \quad (6)$$

U and L are known as the upper and lower boundary sequences. Their positional relationship with the query sequence Q is illustrated in Fig. 4. Obviously, Q is enclosed in the region surrounded by U and L , which is called envelope.

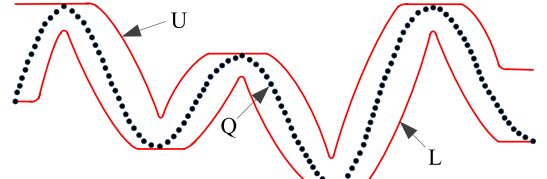


FIGURE 4. The query sequence and its upper and lower boundary sequences.

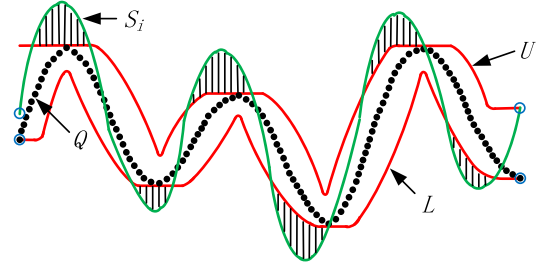


FIGURE 5. Instance of the lower bounding distance LB_Li.

Our lower bounding distance of DTW is defined as:

$$LB_Li(Q, S_i) = |q_1 - s_1| + |q_n - s_n| + \sum_{i=2}^{n-1} \begin{cases} |s_i - U_i|, & \text{if } s_i > U_i \\ |s_i - L_i|, & \text{if } s_i < L_i \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $|q_1 - s_1|, |q_n - s_n|$ are Minkowski distances of (q_1, s_1) and (q_n, s_n) , the rest part is the sum of Minkowski distances between s_i (except s_1 and s_n) and the envelope, as shown in Fig. 5. Since Minkowski distance is calculated n times in Eq. (7), we approximate the computational cost of $LB_Li(Q, S_i)$ by n .

Compared with LB_Keogh, our proposed lower bounding distance improves the tightness by using the first and last original points of the two sequences to calculate the distance. To ensure LB_Li does not introduce false dismissals, we will present Theorem 1 and give a simple proof for it.

Theorem 1: Given two sequences Q, S_i of the same length, and the global constraint on the warping path w_k is $j - r \leq i \leq j + r$, then the following inequality holds

$$LB_Li(Q, S_i) \leq D_{dtw}(Q, S_i) \quad (8)$$

Proof 1: We assume that the cost of the warping path $w_k = (i, j)_k$ is the lowest of all warping paths. According to Eq. (4), we have

$$D_{dtw}(Q, S_i) = \sum_{k=1}^K d(w_k) \quad (9)$$

Then $LB_Li(Q, S_i) \leq D_{dtw}(Q, S_i)$ is equivalent to:

$$|q_1 - s_1| + |q_n - s_n| + \sum_{i=2}^{n-1} \begin{cases} |s_i - U_i|, & \text{if } s_i > U_i \\ |s_i - L_i|, & \text{if } s_i < L_i \\ 0, & \text{otherwise} \end{cases} \leq \sum_{k=1}^K d(w_k) \quad (10)$$

From Eq. (3), we know $\max(n, m) \leq K < n + m - 1$, then $n \leq K$.

According to the constraints of continuity and monotonicity, every point of S_i at least matches a point of Q in the warping path W , as illustrated in Fig. 2(c). For every term on the left-hand side of Eq. (10), we can use a unique term on the right-hand side to correspond to it, leaving $K - n$ terms unmatched. So we have

$$|q_1 - s_1| + |q_n - s_n| + \sum_{i=2}^{n-1} \begin{cases} |s_i - U_i|, & \text{if } c_i > U_i \\ |s_i - L_i|, & \text{if } c_i < L_i \\ 0, & \text{otherwise} \end{cases} \leq \sum_{\text{matched}} d(w_k) + \sum_{\text{unmatched}} d(w_k) \quad (11)$$

Because the warping path W is subject to the boundary conditions and the global constraint r , every term on the left-hand side is not bigger than the corresponding matched term on the right-hand side. We have

$$|q_1 - s_1| + |q_n - s_n| + \sum_{i=2}^{n-1} \begin{cases} |s_i - U_i|, & \text{if } s_i > U_i \\ |s_i - L_i|, & \text{if } s_i < L_i \\ 0, & \text{otherwise} \end{cases} \leq \sum_{k \in \text{matched}} d(w_k) \quad (12)$$

Meanwhile, we know

$$\sum_{\text{unmatched}} d(w_k) \geq 0 \quad (13)$$

Therefore, Eq. (10) holds. Thus, we have $LB_Li(Q, S_i) \leq D_{dtw}(Q, S_i)$.

According to the lower bounding theorem [26], if LB_Li is used for similarity search, the result can guarantee no false dismissals. The relevant proof can refer to [27].

B. EARLY ABANDONING STRATEGY

When calculating the DTW distance of two sequences $Q = q_1, q_2, \dots, q_n$ and $S_i = s_1, s_2, \dots, s_m$, we usually build a cumulative distance matrix $M \in \mathbb{R}^{m \times n}$. Every element of M is defined as:

$$\gamma_{i,j} = D_{base}(q_i, s_j) + \min\{\gamma_{i,j-1}, \gamma_{i-1,j}, \gamma_{i-1,j-1}\} \quad (14)$$

These elements are calculated in a certain direction by rows (or columns) until $\gamma_{m,n}(D_{dtw}(Q, S_i) = \gamma_{m,n})$ is obtained, as shown in Fig. 6.

Then we compares $\gamma_{m,n}$ with the threshold ϵ . Only if $\gamma_{m,n} \leq \epsilon$, S_i is put into the result set $\mathbf{R}$. When there is no global constraint, all the elements $\gamma_{i,j}$ ($1 \leq i \leq m, 1 \leq j \leq n$) in the cumulative distance matrix have to be calculated, as shown in Fig. 6(a). We approximate the computational cost of DTW without global constraint by $m \times n$.

When the global constraint is considered, all the elements $\gamma_{i,j}$ in the Sakoe-Chiba band should be calculated, and its computational cost is approximate by the number of the elements in the band, as shown in Fig. 6(b).

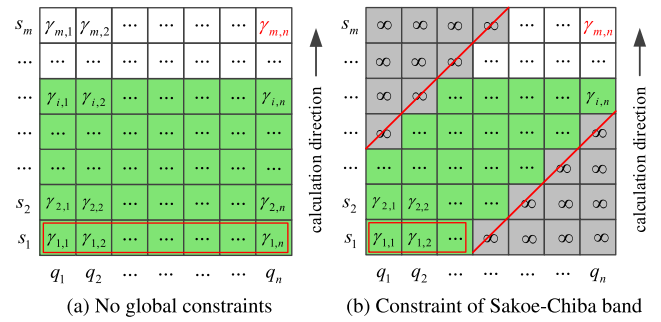


FIGURE 6. Computation process of the DTW distance.

Algorithm 2 The Proposed Two-Stage Similarity Search

Require: Q, S, ϵ ;

Ensure: $\mathbf{R}$;

- 1: foreach sequence S_i in a time series data set $\mathbf{S}$
- 2: if $LB_Li(Q, S_i) \leq \epsilon$ and $LB_Li(S_i, Q) \leq \epsilon$
- 3: add S_i to $\mathbf{C}$;
- 4: end
- 5: foreach sequence S_j in the candidate set $\mathbf{C}$
- 6: if $EA_DTW(Q, S_j) \leq \epsilon$
- 7: add S_j to $\mathbf{R}$;
- 8: end
- 9: return $\mathbf{R}$

Early abandoning strategy can be described as follows [22]. When calculating $\gamma_{i,j}$ in the cumulative distance matrix, once every element $\gamma_{g,j}$ ($1 \leq j \leq n$) in the g -th row is larger than ϵ , according to Eq. (14), we can infer that $\gamma_{m,n}(D_{dtw}(Q, S_i))$ is bound to be larger than ϵ . Thus, there is no need to calculate the rest elements $\gamma_{i,j}$ ($g < i \leq m, 1 \leq j \leq n$) in the cumulative distance matrix.

C. THE PROCEDURE OF THE PROPOSED METHOD

In this subsection, we propose a framework of two-stage similarity search by combining LB_Li and early abandoning strategy, as shown in Algorithm 2.

In the first stage, plenty of the series are discarded by LB_Li , to obtain the candidate set. We need to notice that $LB_Li(Q, S_i)$ and $LB_Li(S_i, Q)$ are respectively compared with ϵ in the second step of Algorithm 2. Fig. 7 illustrates the difference between $LB_Li(Q, S_i)$ and $LB_Li(S_i, Q)$. For $LB_Li(X, Y)$, we always get the envelope of the first sequence X , and then calculate the Minkowski distance between the envelope and the second sequence Y . Since $LB_Li(Q, S_i)$ is usually not equal to $LB_Li(S_i, Q)$, only if the values of $LB_Li(Q, S_i)$ and $LB_Li(S_i, Q)$ are both less than ϵ , dose the sequence S_i enter the candidate set.

In the second stage, early abandoning strategy is used to get the result set from the candidate set. Since the full calculation of DTW is avoided, the proposed method has the potential to improve the efficiency of similarity search. Because both LB_Li and early abandoning strategy will not miss any sequence that satisfies the similar condition, the

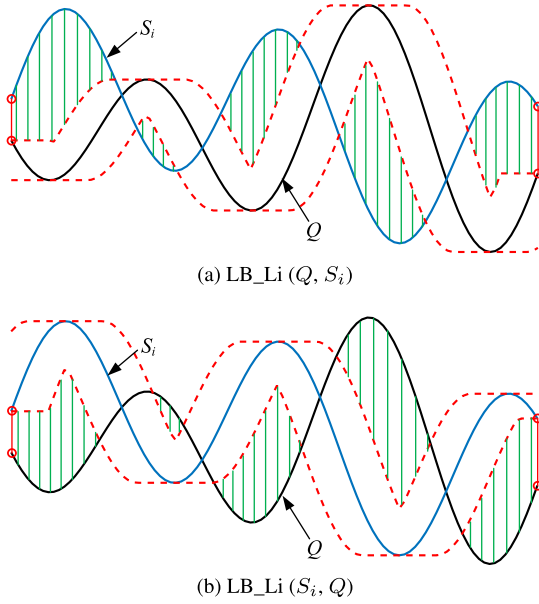


FIGURE 7. Different orders of the sequences in LB_Li.

proposed two-stage similarity search can guarantee no false dismissals.

IV. EXPERIMENTAL EVALUATION

In this section, we implement experiments to verify the effectiveness of our proposed method. We first describe the experimental design, including the experimental setting and procedures. Then we conduct extensive comparisons with competing techniques on real-world data sets. Finally, we discuss the effect of changing the global constraint r . For simplicity, our proposed two-stage method is denoted as LB_Li + EA_DTW.

A. EXPERIMENTAL DESIGN

The experiments are carried out on a PC with Intel(R) Xeon(R) CPU and 8 GB-RAM, running with Matlab R2014a. We compare the computational cost of the proposed method with two competing methods: sequential scan under DTW distance (denoted as DTW) and early abandoning strategy for the calculation of DTW (denoted as EA_DTW). Their computational cost can be evaluated by the methods described in Section III-A.3 and Section III-B.

The experimental process can be described as follows. For a given data set, we respectively take the first five series as the query sequence, and execute similarity search five times independently. We use the mean computational cost to measure the retrieval efficiency of different methods. To make the comparison more intuitive, we take the mean computational cost of DTW as the baseline, and calculate the ratio of the mean computational cost of other methods to the baseline.

Experiments are conducted on 10 real-world data sets from the UCR time series repository [28], including four image data sets, i.e., Yoga, ArrowHead, DiatomSizeReduction, and

TABLE 2. Summary of the data sets.

ID	Data set	Size	Length	Category
1	Yoga	3300	426	2
2	ArrowHead	3300	426	2
3	DiatomSizeReduction	322	345	4
4	Words50	905	270	50
5	Beef	60	470	5
6	GunPoint	200	150	2
7	Car	120	577	4
8	Earthquakes	461	512	2
9	Lighting2	121	637	2
10	ECG5000	5000	140	5

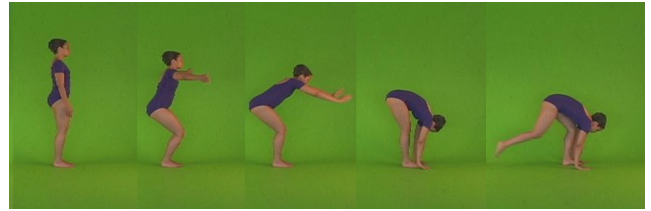


FIGURE 8. Description of Yoga data set.

Words50, one spectrogram data set, i.e., Beef, one motion data set, i.e., GunPoint, three sensor data sets, i.e., Car, Earthquakes, and Lighting2, one ECG data set, i.e., ECG5000. The details of these data sets are shown in Table 2.

B. EXPERIMENTS ON YOGA DATA SET

In our experiments, the global constraint r of the Sakoe-Chiba band equals 10 percent of the length of sequences, consistent with most applications [29].

We first execute experiments on Yoga data set. Yoga is obtained by capturing two actors (a male and a female) transiting between yoga poses in front of a green screen, as illustrated in Fig. 8. Each image was converted to a one dimensional series by finding the outline and measuring the distance of the outline to the centre. The data set contains 3300 instances and the lengths of them are all 426.

The experimental results on Yoga are illustrated in Fig. 9. We can see when ϵ is small, the computational cost of both LB_Li + EA_DTW and EA_DTW is obviously lower than that of DTW. With the increase of ϵ , their computational cost gradually increases, approaching to that of DTW. We summarize the reasons as follows.

For EA_DTW, when ϵ is small, it is more likely to terminate the DTW calculation ahead of time. Therefore, fewer elements of the cumulative distance matrix have to be calculated, which can effectively reduce its computational cost. With the increase of ϵ , more elements of the cumulative distance matrix will be calculated, making the retrieval efficiency decreases gradually. Particularly, if ϵ is large enough, all the elements of the cumulative distance matrix have to be

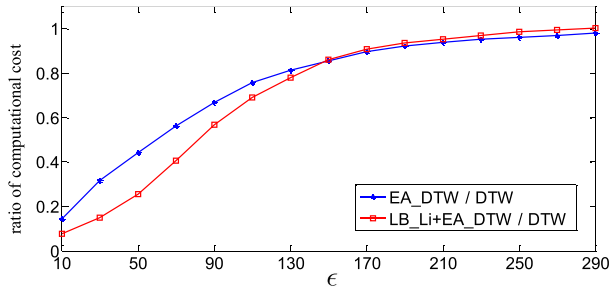


FIGURE 9. Comparison of retrieval efficiency on Yoga.

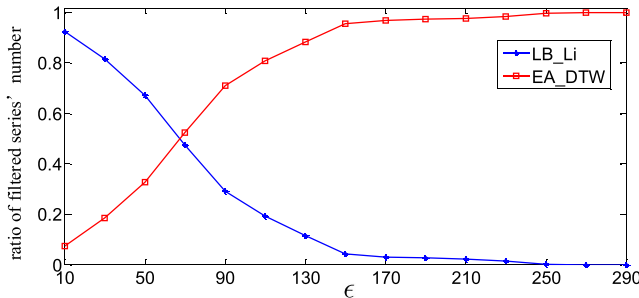


FIGURE 10. The number of series processed by LB_Li or EA_DTW.

calculated. At this time, early abandoning strategy does not work at all and EA_DTW degenerates to DTW.

For LB_Li + EA_DTW, when ϵ is small, parts of dissimilar series are excluded by LB_Li, the rest sequences are processed by EA_DTW. Generally speaking, the computational cost of both LB_Li and EA_DTW is lower than that of DTW. So the computational cost of LB_Li + EA_DTW is much lower than that of DTW. With the increase of ϵ , the filtering effect of LB_Li and EA_DTW is gradually weakened. Therefore, the retrieval efficiency of LB_Li + EA_DTW decreases gradually.

In Fig. 9, there is an interesting phenomenon. When $\epsilon < 150$, the computational cost of LB_Li + EA_DTW is lower than that of EA_DTW. Nonetheless, when $\epsilon > 150$, the former is even slightly higher than the latter. We analyze the reasons as follows.

For EA_DTW, all the sequences are processed by early abandoning strategy. With the increase of ϵ , the computational cost of EA_DTW raises gradually until it reaches that of DTW.

For LB_Li + EA_DTW, when ϵ is small, most dissimilar series are discarded by LB_Li, the computational cost of which is much lower than that of EA_DTW. With the increase of ϵ , the filtering effect of LB_Li is weakened, and more and more series are processed by EA_DTW. Fig. 10 illustrates the number of series processed by LB_Li or EA_DTW.

We can see from Fig. 10, when $\epsilon > 150$, only a few series are filtered out by LB_Li, and almost all the series are processed by EA_DTW. But the calculations of both LB_Li and EA_DTW are inevitable for most of the series. Therefore, when $\epsilon > 150$, the computational cost of LB_Li + EA_DTW is

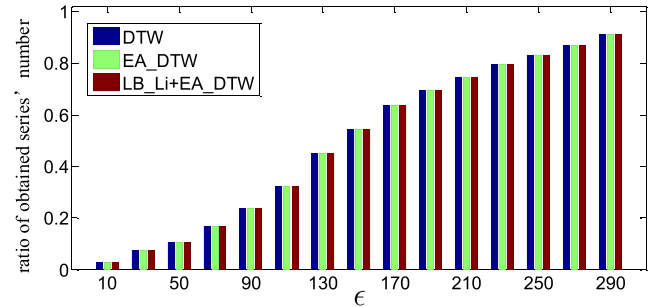


FIGURE 11. The number of series retrieved by different methods on Yoga.

higher than that of EA_DTW. That is, for the whole sequence range search, adding multiple filtering methods, just like the UCR suite, does not necessarily improve retrieval efficiency. The reason is that, if the tightness of filtering methods is too low, it will not discard plenty of sequences, but increases the computational cost.

In order to validate the proposed method can guarantee no false dismissals, we count the number of the series searched by the three methods, and calculate the ratio of this number to the total number of the series, as illustrated in Fig. 11. We can see under different ϵ , the number of the series searched by LB_Li + EA_DTW is same with those of the other two methods. It indicates that our proposed method can find all the series that meet the condition of similarity search.

At this point, you may still worry about the fact that the proposed method will fail when ϵ is large enough. Fortunately, similarity search usually needs to find a small number of sequences from a data set. In Fig. 11, when $\epsilon = 150$, more than 50 percent of all the series are obtained, which can meet the need of most applications. When $\epsilon < 150$, the computational cost of LB_Li + EA_DTW is the lowest. Therefore, compared with DTW and EA_DTW, LB_Li + EA_DTW has higher retrieval efficiency on Yoga data set.

C. EXPERIMENTS ON OTHER DATA SETS

We further extend our experiments from Yoga to other 9 data sets. The experimental results are illustrated in Fig. 12, where the computational cost and the number of the retrieved series are shown in a graph. On all the data sets, the number of the series searched by LB_Li + EA_DTW is always the same with those of the other two methods, which indicates that the proposed method can guarantee no false dismissals. For simplicity, we only show the number of series searched by LB_Li + EA_DTW.

We can see the comparisons are analogous to the result on Yoga. When ϵ is small, the computational cost of LB_Li + EA_DTW is lower than that of DTW and EA_DTW. With the increase of ϵ , the computational cost of LB_Li + EA_DTW is greater than that of EA_DTW, or even greater than that of DTW. The reason is that, with the increase of ϵ , the lower bounding distance LB_Li cannot discard plenty of sequences. Instead, it will increase the computational burden.

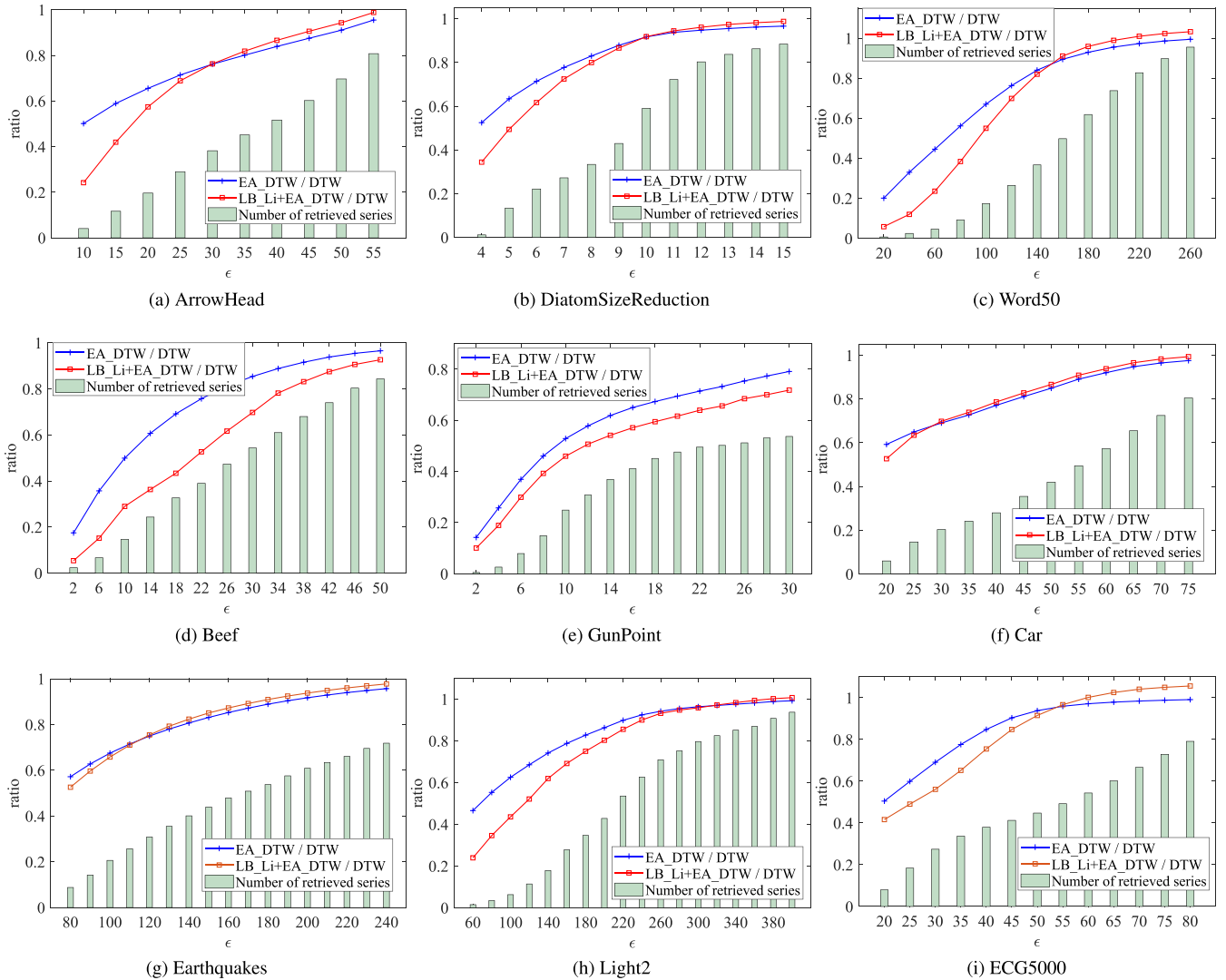


FIGURE 12. Experimental results on other data sets.

Fortunately, before the computational cost of the proposed method exceeds that of DTW and EA_DTW, the number of the searched series is already more than 20 percent of the total number on these data sets, which can meet the need of most applications. Particularly on some data sets, i.e., Beef, and Light2, when the computational cost of LB_Li + EA_DTW is the lowest, the number of the retrieved sequences can reach 80 percent of the total number.

D. SENSITIVITY ANALYSIS FOR GLOBAL CONSTRAINT

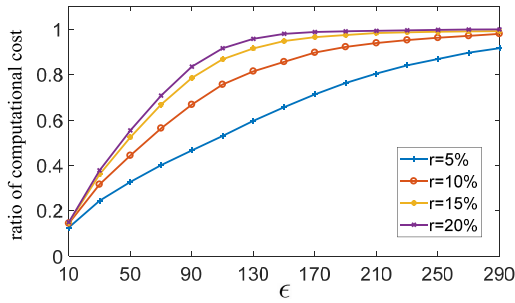
We repeat the above experiments with different global constraint r of the Sakoe-Chiba band. The experimental results on Yoga are illustrated in Fig. 13. We can see, with the increase of r , the computational cost of both LB_Li + EA_DTW and EA_DTW increases gradually.

For EA_DTW, with the increase of r , the Sakoe-Chiba band becomes wider, as shown in Fig. 6(b). More elements in

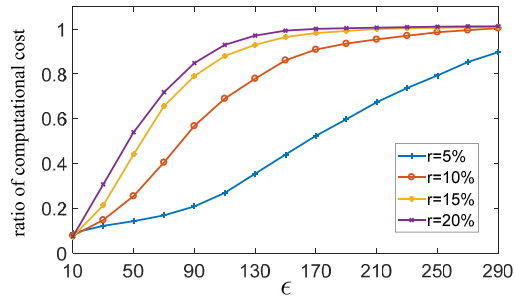
the cumulative distance matrix have to be calculated. Besides, the condition of early abandoning strategy is more difficult to satisfy. Thus, the computational cost of EA_DTW rises gradually.

For LB_Li + EA_DTW, with the increase of r , LB_Li of the two sequences is bound to decrease. An instance of LB_Li is illustrated in Fig. 14, where S_1 , S_2 are the first two sequences of Yoga, coming from different categories. $U(S_1)$, $L(S_1)$ respectively denote the upper and lower boundary sequences of S_1 . The shadow parts represent LB_Li(S_1 , S_2).

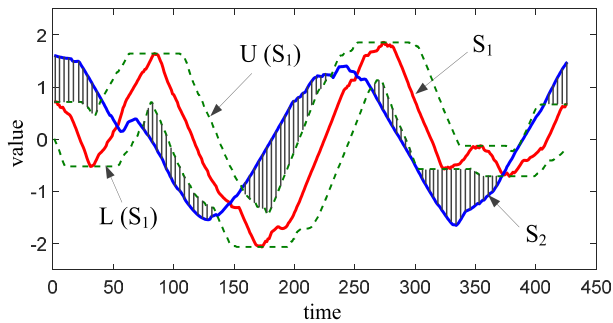
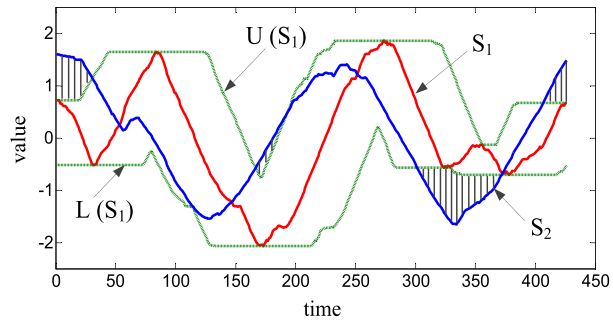
For a given r , LB_Li of the two sequences is a constant. When r changes from 5 percent to 10 percent of the length of sequences, the area of the shadow parts decreases sharply, as shown in Fig. 14. Then the filtering effect of LB_Li will become weaker, and more series are processed by subsequent EA_DTW. So the computational cost of LB_Li + EA_DTW rises with the increase of r .



(a) The computational cost of EA_DTW



(b) The computational cost of LB_Li + EA_DTW

FIGURE 13. The effect of changing r on Yoga.(a) An instance of LB_Li(S_1, S_2) when $r=5\%$ (b) An instance of LB_Li(S_1, S_2) when $r=10\%$ **FIGURE 14.** The instances of LB_Li under different r .

V. CONCLUSION

In this paper, we propose a simple framework for the whole sequence range search, by combining lower bounding distance and early abandoning strategy together. Then, we propose an improved lower bounding distance LB_Li,

and complete relevant proof that LB_Li can guarantee no false dismissals. Finally, extensive experiments are conducted on real-world data sets. The experimental results verify the effectiveness of our proposed method.

Several aspects need to be studied further, for example, how to speed up the retrieval efficiency of multivariate time series under DTW is a promising research direction.

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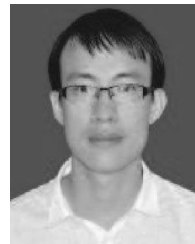
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